

ALEK FRÖHLICH

Ph.D. Student in Machine Learning

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EDUCATION

ELLIS Ph.D. in Computer Science

Istituto Italiano di Tecnologia & Institut Polytechnique de Paris

Nov 2024 – Present

Genoa, Italy

- Research: Representation learning and spectral methods for nonparametric statistical inference, with a focus in causality and uncertainty quantification.
- Advisors: [Massimiliano Pontil](#), [Karim Lounici](#), [Vladimir Kostić](#).

M.Sc. in Applied Mathematics

Federal University of Santa Catarina

Mar 2022 – Jul 2024

Florianópolis, Brazil

- GPA: 9.76/10. Courses: Functional analysis, machine learning theory, numerical optimization, differential equations, topology, advanced calculus, linear algebra.

B.Sc. in Computer Science

Federal University of Santa Catarina

Mar 2018 – Mar 2022

Florianópolis, Brazil

- GPA: 9.23/10 (2nd in cohort). Some courses: Measure theory, machine learning, probability, statistics, parallel programming, algorithms, software engineering, operating systems.

Exchange Student

University High School

Aug 2016 - Dec 2016

Irvine, CA

- AGPA: 4.2. AP Course: Computer science.

EXPERIENCE

Research Intern

Federal University of São Carlos

Jul 2024 – Sep 2024

São Carlos, Brazil

- Developed an interpretable breast cancer risk assessment tool with local coverage uncertainty quantification guarantees via conformal prediction. Resulted in [2]. Mentor: [Rafael Izbicki](#).

Research Intern

University of São Paulo Medical School

Jul 2023 – Jul 2024

Ribeirão Preto, Brazil

- Developed and validated machine learning models for predicting malignancy of suspicious breast lesions identified during ultrasound examinations. Resulted in [3].
- Developed a computational pathology pipeline for segmenting and classifying tumor cells in whole slide images. Resulted in [4]. Mentor: [Daniel Tiezzi](#).

Visiting Student

Institute for Pure and Applied Mathematics (IMPA)

Jan 2023 – Nov 2023

Rio de Janeiro, Brazil

- Courses: Functional analysis and machine learning theory.
- Seminars: Optimization (mentored by [Benar Svaiter](#)) and physics-informed neural networks.

PUBLICATIONS & PRE-PRINTS

1. D Ordoñez-Apraez, V Kostić, **A Fröhlich**, V Brandt, K Lounici, M Pontil. *Equivariant Representation Learning for Symmetry-Aware Inference with Guarantees*. [\[arXiv\]](#)
2. **A Fröhlich**, T Ramos, G Santos, I Buzatto, R Izbicki, D Tiezzi. *PersonalizedUS: Interpretable Breast Cancer Risk Assessment with Local Coverage Uncertainty Quantification*. AAAI 2025. [\[DOI\]](#)
3. I Buzatto, S Recife, L Miguel, R Bonini, N Onari, A Faim, L Silvestre, D Carlotti, **A Fröhlich**, D Tiezzi. *Machine Learning can Reliably Predict Malignancy of Breast Lesions based on Clinical and Ultrasonographic Features*. Breast Cancer Research and Treatment 2024. [\[DOI\]](#)
4. D Tiezzi, **A Fröhlich**, F Chahud, S Pagnota. *197P Computational pathology pipeline enables quantification of intratumor heterogeneity and tumor-infiltrating lymphocyte score*. ESMO Immunology and Technology 2023. [\[DOI\]](#)

TALKS

Spectral Representation Learning <i>University of Genoa</i>	Oct 2025 Genoa, Italy
Spectral Representation Learning for High-dimensional Inference <i>Federal University of São Carlos</i>	Aug 2025 São Carlos, Brazil
Breast Cancer Risk Assessment with Local Coverage UQ <i>New Trends in Statistical Learning V</i>	Jun 2025 Porquerolles, France
Machine Learning Methods in Her2+ Breast Cancer <i>Institute for Pure and Applied Mathematics (IMPA)</i>	Oct 2023 Rio de Janeiro, Brazil
AI and Her2+ Breast Cancer <i>University of São Paulo Medical School</i>	Aug 2023 Ribeirão Preto, Brazil
SVM: Optimization Problem, Learning Guarantees, and Kernel Trick <i>34th Brazilian Mathematical Colloquium</i>	Jul 2023 Rio de Janeiro, Brazil

SUPERVISIONS

Enzo N. Spotorno (co-advised with A. A. Fröhlich) <i>Federal University of Santa Catarina</i> Project: Improving Battery End-of-Life Prediction with Hybrid Bayesian PINNs	Jul 2024 – Jul 2025
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SCHOLARSHIPS

EVENTS

Mathematical Aspects of Data Science Summer School <i>École Polytechnique Fédérale de Lausanne (EPFL)</i>	Sep 2025 Lausanne, Switzerland
Theoretical Foundations of Machine Learning Summer School <i>University of Genoa</i>	Jun 2025 Genoa, Italy

New Trends in Statistical Learning V <i>Presented [2]</i>	Jun 2025 Porquerolles, France
39th Annual AAAI Conference on Artificial Intelligence <i>Presented [2]</i>	Feb 2025 Philadelphia, PA
Workshop on Optimization and Inverse Problems <i>Federal University of Santa Catarina</i>	2023 Florianópolis, Brazil
34th Brazilian Mathematical Colloquium <i>Institute for Pure and Applied Mathematics (IMPA)</i>	2023 Rio de Janeiro, Brazil
Machine Learning and Combinatorics Workshop <i>Moscow Institute of Physics and Technology</i>	2020 Moscow (online), Russia
Brazilian Symposium on Computing Systems Engineering IX <i>Federal University of Rio Grande do Norte</i>	2019 Natal, Brazil
The Developer's Conference <i>Largest event related to software development in Brazil</i>	2019 Florianópolis, Brazil
Chip in the Pampa 2018 <i>Federal University of Pelotas</i>	2018 Pelotas, Brazil
Workshop on Quantum Computing I <i>Federal University of Santa Catarina</i>	2018 Florianópolis, Brazil

SKILLS

Languages	Portuguese (native), English (fluent), Italian (basic)
Programming	C/C++, Java, Python, R, Matlab, PyTorch
Miscellaneous	Git, Docker, L ^A T _E X, SQL, Assembly (x86, ARM, RISC-V, MIPS)